

Calcium supplementation and bone mineral density in females from childhood to young adulthood: a randomized controlled trial.

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BACKGROUND: Short-term studies established that calcium influences bone accretion during growth. Whether long-term supplementation influences bone accretion in young adults is not known. **OBJECTIVE:** This study evaluated the long-term effects of calcium supplementation on bone accretion among females from childhood to young adulthood. **DESIGN:** A 4-y randomized clinical trial recruited 354 females in pubertal stage 2 and optionally was extended for an additional 3 y. The mean dietary calcium intake of the participants over 7 y was approximately 830 mg/d; calcium-supplemented persons received an additional approximately 670 mg/d. Primary outcome variables were distal and proximal radius bone mineral density (BMD), total-body BMD (TBBMD), and metacarpal cortical indexes. **RESULTS:** Multivariate analyses of the primary outcomes indicated that calcium-supplementation effects vary over time. Follow-up univariate analyses indicated that all primary outcomes were significantly larger in the supplemented group than in the placebo group at the year 4 endpoint. However, at the year 7 endpoint, this effect vanished for TBBMD and distal radius BMD. Longitudinal models for TBBMD and proximal radius BMD, according to the time since menarche, showed a highly significant effect of supplementation during the pubertal growth spurt and a diminishing effect thereafter. Post hoc stratifications by compliance-adjusted total calcium intake and by final stature or metacarpal total cross-sectional area showed that calcium effects depend on compliance and body frame. **CONCLUSIONS:** Calcium supplementation significantly influenced bone accretion in young females during the pubertal growth spurt. By young adulthood, significant effects remained at metacarpals and at the forearm of tall persons, which indicated that the calcium requirement for growth is associated with skeletal size. These results may be important for both primary prevention of osteoporosis and prevention of bone fragility fractures during growth.